

Defectivity and auxiliary syncretism: diachronic aspects*

Dalina Kallulli¹ and Ian Roberts²

University of Vienna¹

University of Cambridge²

Abstract

In this article, we argue for the existence of neutralised (i.e. featurally underspecified) auxiliaries, and consequently that the feature content of auxiliaries can change over time in such a way that it interacts with the labelling algorithm. Our core contention is that auxiliaries which are underspecified for being HAVE or BE, such as English *ain't* and Southern Italo-Romance *seva/sɔva*, have an optional LOC(ATIVE) feature. For example, while 's in a sentence like *There's students in the room* is a neutralised auxiliary with an optional LOC feature, which crucially involves LOC agreement between *there* and (the neutralised) 's, and a sentence like *There are students in the room* involves ϕ -agreement between *there* and *are*, the non-contracted *is* in the ungrammatical **There is students in the room* is unambiguously a form of *be* and so lacks a LOC feature. As such, it involves neither LOC agreement between *there* and neutralised 's, or ϕ -agreement between *there* and *are*, thus leaving the root node unlabelled and leading to ungrammaticality. We argue that non-neutralised HAVE-auxiliaries on the other hand have an obligatory LOC feature. By comparing the development of the two different cases of neutralised auxiliaries, we see that the diachronic convergence of HAVE and BE can arise in different ways (phonologically or morphologically). We also show that our account of the neutralised auxiliaries can readily be integrated into existing approaches to locatives and existentials.

* Versions of this article were presented at DiGS 23 (NYU) and the DGfS 2023 Workshop 'Economy, routine, and syntactic change' (University of Cologne). We are grateful to these audiences for questions and comments, to two anonymous reviewers, and to Ailis Cournane, David Denison, Markus Pöchtrager and Gary Thoms for invaluable feedback.

1. Introduction

Consider the following paradigm in Modern Standard English from Kallulli (2008: 286):¹

- (1)a. There are students in the room.
- b. *There is students in the room.
- c. There's students in the room.

Kallulli suggests that 's is really a reduced form of *has*, comparable to French *il y a*, Spanish *hay*, etc., as in (2):²

- (2)a. *Il y a des étudiants dans la salle.* (French)
3SG,M LOC have.3SG some students in the room
- b. *Hay estudiantes en la sala.* (Spanish)
there-is students in the room
'There are students in the room.'

Here we propose that 's in (1c) is a neutralised HAVE/BE auxiliary, with an optional locative (henceforth: LOC) feature, while non-neutralised HAVE-auxiliaries have an obligatory LOC feature (see Benveniste 1960, Freeze 1992, Den Dikken 1995).³ Non-contracted *is* on the other

¹ All acceptability judgements on (Modern) English examples reflect Roberts's native speaker intuitions (a speaker of conservative Northern-influenced Standard British English). As a reviewer notes, of course there's variation in the English-speaking world in this regard, as in others. We comment on this variation as appropriate in what follows.

² A reviewer points out the ungrammaticality of (i):

(i) *There're a student in your office.

Here the reduced auxiliary, like its full-form counterpart in (1a), has ϕ -features which agree with *there*, unlike the neutralised 's in (1b). In some varieties of Scots, forms such as *We're to go* and *We're got the time* are found (thanks to Gary Thoms, p.c., for this observation); this might be a further case of a neutralised auxiliary.

³ As a reviewer points out, Kayne (1993, 2000) on the other hand makes reference to an abstract D/P-element rather than a locative feature, which he moreover does not take to be locative. However, since the pioneering work of Benveniste, the connection between location and possession has been recognized (for discussion, see Hallman 2022). Our use of the LOC feature is intended to capture this intuition. For our purposes here, we leave aside the question of whether there might be finer-grained distinctions between location and (certain types of) possession (see Boneh and Sichel 2010, Levinson 2011, Hallman 2022:564), noting however that even a preposition like *with* in English (as in: *a man with a hat*) can be locational (the hat is on the man), contra Levinson (2011). We thus disagree with Kayne (1993, 2000) and Levinson (2011) on the non-locative source of HAVE-auxiliaries.

The same reviewer correctly notes that the English varieties that allow 'There's students in the room' also seem "to allow 'Where's all the students?', 'Where's my glasses?', with a non-agreeing 's. We see no bar to treating examples of this kind as simply involving the wh-counterpart of *there*, i.e. *where*, which therefore has a LOC feature agreeing with 's but not *is* and a wh-feature agreeing with [+wh] C after string-vacuous movement to SpecCP (exactly as in *Who left?*). Examples with *where* in situ such as **The children's where?* also pointed out

hand is unambiguously a form of *be* and as such lacks a LOC feature altogether. Following a large body of research that takes *there* to be the subject of predication (Jenkins 1975, Williams 1994, Hazout 2004, Kallulli 2008, a.o.), we can then account for the paradigm in (1) as follows: (1a) involves ϕ -agreement between *there* and *are*; (1c) involves LOC agreement between *there* and neutralised 's, and (1b) involves neither, leaving the root node unlabelled, thus leading to ungrammaticality.

In the rest of this article, we will flesh out this proposal in more detail. In section 2, we outline the nuts and bolts of our proposal, provide a fuller picture of the English data in their diachronic development, and then turn to a discussion of Italo-Romance varieties where an analogous HAVE/BE auxiliary neutralisation can be observed. In section 3, we then broaden the discussion and investigate the relationship between locative and possessive constructions in more general terms, following the influential proposals of Freeze (1992) and Den Dikken (1995), before wrapping up with the conclusion that labelling drives change (section 4).

2. Analysis

2.1. Core proposal

The crux of our proposal to account for the paradigm in (1) is given in (3):

- (3)a. Non-neutralised HAVE-auxiliaries have **obligatory** LOC (Benveniste 1960, 1966, 1971, Freeze 1992, Den Dikken 1995)
- b. 's is a neutralised HAVE/BE auxiliary, which we capture by attributing an **optional** LOC feature to it
- c. Non-neutralised BE-auxiliaries have **no** LOC

Thus, in (1a), there is ϕ -agreement (for Number), with the root labelled ϕ , as depicted in (4a). In (1b), there is LOC-agreement, with the root labelled LOC, as in (4b). In (1c) on the other hand, there is failure of agreement since *is* does not have a LOC feature and the 3SG features of *is* are unspecified, realized as 3SG by default. If, in the case of the XP YP labelling configuration of

by the reviewer, are ungrammatical owing to a failure of labelling (at the “TP”, not the “CP”, level, since *the children* and 's do not share features. The ungrammaticality of ‘*Where have all the students?’, ‘*Where have my glasses?’ follows from the account of (9) in section 2.1, if *where* is just the wh-counterpart of *there*. Indeed, the fact that ‘*There has students in the room’ is ungrammatical might at first seem to be unexpected under our analysis given the obligatory LOC feature on HAVE-auxiliaries. We discuss this issue in detail in section 2.1.

Chomsky (2013, 2015), both XP and YP must share the labelling feature, then (4c) results in a failure of labelling and for this reason is ungrammatical.

- (4)a. There_φ are_φ students in the room.
- b. There_{LOC} 's_{LOC} students in the room.
- c. *There_{LOC} is students in the room.

This raises the question of the well-formedness of (5):

- (5) There is a student in the room.

The difference between (5) and (4c) is that there is associate agreement in number in (5) but not in (4c). Whatever the precise technical details, associate agreement must involve feature sharing between the copula and the associate, hence in (5) the 3SG features on *is* are non-default. The same is true for (6):

- (6) There's a student in the room.

The contracted form *'s* in (1) does not seem to correspond to a full form *is* (unlike the contracted form in (6)). This is confirmed by the following contrasts in inversion, negation, emphatic forms, and under VP-Ellipsis. As we see in (7), *'s* cannot surface as *is* where the associate is plural but it can in (8) where the associate is singular:⁴

- (7)a. Are/*is there students in the room?
- b. There aren't/*isn't students in the room.
- c. There ARE/*IS students in the room.
- d. John said there's students in the room and there *is/*'s/are.

⁴ Thoms et al. (2019) discuss a similar contraction in varieties of Scots English in what they call "LOCATIVE DISCOVERY EXPRESSIONS (LDES)" -- including things like *There he's*, which seemingly violate the restriction against the occurrence of a contraction in the immediate context of a gap created by movement or ellipsis -- and more generally on syntactic (micro-)variation and auxiliary contraction. Compare SE *There he is/*'s*. It is clear that this construction has rather different properties from the one in (1b), which we focus on here. The same can be said for things like *There's the lions* (see Close 2004), and also for examples like *There was just the three of us*, which a reviewer draws our attention to (note also the contrast in *There was / ??is just the three of us*).

- (8)a. Is there a student in the room?
 b. There isn't a student in the room.
 c. There IS a student in the room.
 d. John said there's a student in the room and there is.

We conclude that 's in (1c) is a neutralised HAVE/BE auxiliary, probably the only one in Modern SE. However, once we take into consideration non-standard varieties, we see that there clearly exist further instances of such neutralisations. One case in point is *ain't* in non-standard English, as we discuss in 2.2 and 2.3. A second case involves *seva* in Italo-Romance varieties, as we discuss in section 2.4. In both cases, we look at diachronic developments, as well as the relationship between the auxiliaries HAVE and BE (in section 3), and conclude that labelling drives syntactic change.⁵

Before moving on to these issues, however, we need to consider a potential problem raised by the following example, as we pointed out in note 3. This is the fact that (9) is ungrammatical, which might at first seem to be unexpected under our analysis given the obligatory LOC feature on HAVE auxiliaries:

- (9) *There has students in the room.

In this connection, Freeze (1992:583) observes in a section entitled “Subjects of *have* are [+human]” (his section 4.3) that subjects of *have* are preferentially [+human] in English, and when they are [-human], they are restricted to inalienably possessed arguments and/or require a PP headed by *in* containing an anaphoric pronoun, giving the following examples (his (62a, c, e) p. 583):

- (10) a. The tree has branches.
 b. *The tree has a nest.
 c. The tree_i has a nest in it_i. (= There is a nest in the tree.)
 d. The flour_i has a ring in it_i. (= There is a ring in the flour.)

⁵ The fact that the neutralised auxiliary is a reduced form, is the result of a diachronic reanalysis of this particular form. We do not require that neutralised auxiliaries be reduced in general, and this is clearly not the case, as the discussion of *ain't* and *seva/sova* in the sections to follow shows.

Expletive *there* is obviously [-human], being non-referential. Furthermore, *there* cannot be associated with an anaphoric pronoun because it is non-referential, ruling out the possibility of an analogue to (2c, d). Finally, *there* cannot bear the thematic role associated with inalienable possession (arguably a locative/experiencer role; see Landau 2010), being an expletive. Therefore, *there* cannot appear as the surface subject of *have* predicates,⁶ despite being able to share a LOC feature (which must be distinguished from the locative/experiencer thematic role borne by inalienable possessors) with *have*. Feature-agreement is necessary but not sufficient for grammaticality in this case. English *have* (but not the neutralized ‘s) differs from French *avoir* of the *il y a* construction and Spanish *hay* in (2) in that it cannot select a “pure” existential small-clause complement, but arguably only the small-clause complements of the kind seen in note 6.⁷

More generally, we speculate that the reason that *have* generally imposes a [+human] requirement is that, in its existential/locative usage in English (see note 7 on the other uses of *have*), it is a psych verb akin to the *fear*-type. More precisely, this kind of *have* selects a small-clause predicate headed by an abstract Pred element which assigns an Experiencer thematic role to a [+human] external argument (see also note 6). Corroborating evidence for this comes from other psych-predicates which partially participate in an analogue to the *spray-load* alternation, as the following examples show:

- | | | | |
|------|----|---|-------------------|
| (11) | a. | We loaded the wagon with hay. | Location-Theme |
| | b. | We loaded hay onto the wagon. | Theme-Location |
| | c. | We amused the children with the stories. | Experiencer-Theme |
| | d. | *We amused the stories into the children. | Theme-Experiencer |

The ungrammaticality of (11d) is due to the fact that *the stories* cannot be the Experiencer external argument of the small clause selected by *amuse* here. Our conclusion regarding

⁶ It is very likely that these examples derive from small clauses in which the possession relation is expressed by means of a “subject-predicate” like structure, as in (i):

- (i) The tree has [(the tree) [Pred branches]].
- (ii) The tree has [a nest [in it]].

In (ii) (where we arguably have stage-level possession as opposed to individual-level possession in (i)) the predication relation is “inverted” and the head of the small clause is overt. The pronoun may be required as movement of *the tree* across the small-clause subject would violate relativized minimality. See Thoms et al (2023b) for a more detailed analysis of existential and locative *have* along similar lines.

⁷ Perfect and modal *have* are different from the main-verb *have* seen here, which requires *do*-support in the standard contexts as can be seen from the negations of the examples in (4). Causative *have* as in *I had Jeeves wash the car* differs again in having its own external (Agent) thematic role.

locative/existential *have* is fully consistent with the proposal that Experiencers are locatives with a [+animate/human] feature (see Landau's 2010 and references therein); this then is a lexical property of this kind of *have* subject to the proviso noted by Freeze discussed above.

Let us now move on to look at the other cases of neutralised auxiliaries and their diachronic development.

2.2. The curious case of *there's* and its cousin *ain't*

Non-standard *ain't* is clearly syncretic for *have* and *be*, as shown in (12),⁸ thus contrasting with SE which in these contexts distinguishes for the Number/Person features (i.e. between *haven't* vs. *hasn't* and *aren't* vs. *isn't*, respectively).

- | | | |
|------|--|------------------------|
| (12) | a. I/you/he/we/they ain't got no money. | (SE: haven't / hasn't) |
| | b. I/you/he/we/they ain't going nowhere. | (SE: aren't / isn't) |
| | c. I/you/he/we/they ain't students. | (SE: aren't / isn't) |
| | d. I/you/he/we/they ain't bothered what you think. | (SE: aren't / isn't) |

As is clear from the paradigm in (12), the synchronic features of non-standard *ain't* are: [AUX, NEG, (LOC)].⁹ That is, *ain't* is clearly a negative auxiliary and, being fully syncretic between HAVE and BE, we attribute to it an optional LOC feature. We can also clearly see from (12) that there is no Person/Number specification. So, in the contemporary non-standard dialects where it is found, *ain't* is best synchronically analysed as a negative auxiliary with an optional LOC feature and no Person/Number specification. This has the consequence that, in these English varieties, (13) would have the same analysis as SE (1c) (see (4b)):

- (13) There_{LOC} ain't_{LOC} no students in the room.

Since the LOC feature is shared between *there* and *ain't*, the root is labelled LOC and the sentence is grammatical.

Modulo the restriction to negative auxiliaries, which is clearly linked to contracted negation (see the diachronic account in section 2.3. below), (13) is equivalent to the French and Spanish

⁸ Note that these varieties typically have negative concord.

⁹ We take the feature AUX to be a shorthand for elements merged in (finite) T. See below for a discussion of *ain't* as a syncretic form for main-verb possessive *have*.

examples with a HAVE auxiliary in existential sentences seen in (2). Here too there is a LOC feature, present on the locative clitic *y* in French and on what is diachronically the combination of HAVE and the locative *y* in Spanish. This is shown in (14):

- (14)a. *Il_{LOC} y_{LOC} a des étudiants dans la salle.* (French)
 3SG,M LOC have.3SG some students in the room
- b. *Hay_{LOC} estudiantes en la sala.* (Spanish)
 there-is students in the room
 ‘There are students in the room.’

Spanish is a null-subject language, so in (14b) there may be an expletive *pro_{LOC}* in the highest Spec, or we can assume, following Chomsky (2015), that the “rich” agreement which licenses null subjects in languages like Spanish is “strong” enough to label the root alone. What is clear though is that, in (14), here too the root can be labelled LOC as in (4b) and (13).

With this background, we can now turn to the diachrony of English *ain’t*.

2.3. A brief history of *ain’t*

Diachronically, *ain’t* was initially a contraction of BE (see Cheshire 1982, Lass 1999, and the Wikipedia entry on *ain’t*, which the following data are taken from unless otherwise acknowledged). *Amn’t* as a contraction of *am not* first appears in 1618, it is later written *an’t*, appearing in print in this form in 1695,¹⁰ in the following line from Congreve:

- (15) I can hear you farther off, I an’t deaf. (Congreve *Love for Love*, p. 55)

Aren’t as a contraction for *are not* first appeared in 1675, written as *an’t* first in 1696:

- (16) These shoes an’t ugly, but they don’t fit me. (Vanbrugh *The Relapse*, p. 13)

¹⁰ *Amn’t* still survives as such in some Hiberno-English varieties. See also Broadbent (2009), who argues that contemporary West Yorkshire has never had a **amn’t* gap -- specifically, she argues that “secondary contraction” is responsible for the creation of homophones for *amn’t* and *aren’t*: [a:nt]/[a:t] -- and more generally for the idea that certain AUX+*n’t* forms have become lexicalised and that this has triggered secondary contraction as a phonological repair strategy. She then pursues the possibility that lexicalisation may have occurred in precursors of Standard British English, and that homophony for *amn’t* and *aren’t* may have led to prescription against new realisations of *amn’t*. See also Thoms et al (2023) on *amn’t* and related matters in Scots English.

Like the contracted forms of modals, which first appear in the 17th century (Lass 1999: 180), a phonological rule deletes the final voiced continuant consonant of the auxiliary, with lengthening of the vowel (Lass 1999: 103f., 180):¹¹

- (17) a. [+cons, +cont] → ∅ / __ n't
 b. V → V: / __ n't

This rule gives rise to the following contracted forms of modals which first appear in the 17th century (Lass 1999:103f., 180):

- (18) *shall* > *shan't*
can > *can't*
will > *won't*

Rule (17a) deletes the /l/ of *shall*, the /n/ of *can* (the surviving /n/ is that of *n't*) and the /l/ of the older stem form *wol* (Lass 1999:178). The vowels are lengthened in accordance with rule (17b); in the case of *won't* there is further diphthongisation.

The development of *amn't* to *an't* and *aren't* to *an't* falls into this general pattern. Rule (17a) deletes the /m/ of *amn't* and the /r/ of *aren't* (in what was at the time the majority rhotic dialect). In the 3rd person the change from *isn't* to *in't* or *en't* obeys rule (17a) in deleting /z/. There appears to be a later change in vowel quality, judging from later *an't*. This form appears for *isn't* in Swift in 1710–13:

- (19) *It an't my fault, 'tis Patrick's fault; pray now don't blame Presto*
 (Swift, *Journal to Stella*, Letter 19)

The /a/ vowel is lengthened in accordance with (17b) and then diphthongised, giving *ain't*. This form is first attested in writing in 1749. In this way, all the negative present-tense forms of BE result as *ain't*.

¹¹ Rule (17a) predicts deletion of /z/ also in *hasn't*, *doesn't* and *wasn't*. This is actually correct for *hasn't*, as we will see below. In many non-standard varieties of English, the rule has also applied to *doesn't* and *wasn't*, yielding *don't* and *weren't* in these varieties. The fact that *hasn't*, *wasn't* and *doesn't* survive in Modern SE but *shalln't* and *willn't* do not, suggests that in SE (17a) should be replaced by (i):

i. [+cons, -cont, +son] → ∅ / __ n't

As formulated, rule (i) will only apply to laterals and nasals but not to /z/. Thanks to Markus Pöchtrager (p.c.) for helping us formulate this phonological rule.

The phonological rules in (17) also apply to negative forms of HAVE. Thus, forms without /s/ in the 3rd person and without /v/ in other persons appear in the late 17th century. So, we observe *han't/ha'n't* for *has not* and *have not* from 1675:

- (20) *Gentlemen and Ladies, han't you all heard the late sad report / of poor Mr. Horner.*
(Wycherly *The Country Wife*)

Vowel-lengthening as in (17b) and diphthongisation give *hain't*. Finally, dropping of initial /h/, found in almost all non-standard varieties of English, gives *ain't*, which first appeared as a contraction of *have not* in print in 1819:

- (21) *Strike! Why I ain't got nobody here to strike...* (*Niles' Weekly Register* Vol. 16. p.190)

So, we see that morpho-phonological processes, the rules in (17) combined with diphthongisation and, in the case of HAVE, /h/-dropping, caused the two forms of the negative auxiliaries to converge as a single neutralised auxiliary by the early 19th century. This neutralised auxiliary has an optional LOC feature, as we saw. It is intrinsically negative, although this is a contingent fact related to the environment for the rules in (17a) which, as we have just seen, also gave rise to the negative forms of modals. Finally, again like the modals, *ain't* has no Person- or Number-agreement features. Thus, the culmination of the diachronic processes is *ain't* with the features [AUX, NEG, (LOC)], as we saw above.¹²

The example in (21) illustrates two further relevant points. First, as already pointed out, the varieties of non-Standard English which have *ain't* uniformly have negative concord, as far as we are aware. Second, (21) is a case of the “have-got”-possessive construction, which began replacing main-verb possessive *have* in British English from the late 18th century (Roberts

¹² A reviewer points out that there are varieties of American English in which *ain't* is syncretic for BE but not HAVE. This is consistent with the history of *ain't* as just described, in that we can postulate that these varieties represent a more conservative grammar in which HAVE has not converged with BE, corresponding to an earlier stage of at least Modern British English as described in the text (see also note 1). As the same reviewer also remarks, “not all speakers of English have *ain't* at all, even in their non-standard variety (e.g. in Australia)”. In our view, this may be the consequence of a strong normative pressure against the use of *ain't*, a view which is further corroborated by the existence of examples such as the one in (i) below apparently from Australian English, found in the online Oxford English Dictionary (‘Quotation Text’ under *ain't* <https://www.oed.com/search/advanced/Quotations?textTermText0=ain%27t&textTermOpt0=QuotText&dateOfUseFirstUse=false&page=1&sortOption=AZ>):

i. ‘Starve the lizards,’ he said, ‘there ain’t no kangaroos in the West now’. *Bulletin* (Sydney) (1927)
Conversely, in African American English *ain't* appears in contexts where SE has *didn't* (see Fisher 1998); thanks to Gary Thoms for drawing our attention to this.

1993:340). It is for this reason that forms such as *They ain't no money*, which a reviewer brings to our attention as a potential problem, are very rare. Main-verb *have* was replaced by *have got* at approximately the same time as *ain't* became syncretic for *haven't/hasn't*. This is almost certainly a case of pure historical contingency; two examples of possessive *ain't* were found in the online *Oxford English Dictionary*:¹³

- (22) a. You ain't no idee what a getting' up bluffs, and general absentin' of 'emselves ther wur.
W. E. Webb, *Buffalo Land* (1872)
- b. 'You ain't no idee', said he, 'how strong the arch is if ye set it right'.
Harper's Magazine (1882)

Both examples have *no idea* as the object of *ain't*. It is perhaps worth noting that this is a context which favours main-verb *have* in Modern British SE.

Having seen evidence for HAVE/BE auxiliary neutralisation in the history of non-standard English, in the next section we broaden the scope of inquiry and the appeal of our analysis by adducing corroborating data for exactly the same process of HAVE/BE auxiliary neutralisation from Italo-Romance varieties.¹⁴

2.4. Neutralised HAVE-BE auxiliaries in Italo-Romance (Cennamo 2010)

Cennamo (2010) discusses neutralised HAVE/BE auxiliaries in Southern Italo-Romance, more precisely in certain Campanian and Molisan dialects. In these varieties, the HAVE/BE alternation in the perfect is typically controlled by Person. In other words, the auxiliary is HAVE or BE in a single tense as a function of the Person-Number of the subject (for extensive discussion and illustration, see Loporcaro 2016, Manzini and Savoia 2005, II: 649-654; III: 1-34). In the pluperfect of the relevant Campanian and Molisan dialects a form *seva* appears, often

¹³ <https://www.oed.com/search/advanced/Quotations?textTermText0=ain%27t&textTermOpt0=QuotText&dateOfUseFirstUse=false&page=1&sortOption=AZ&tl=true> (last accessed August 8, 2023)

¹⁴ Auxiliary neutralisation also extends to *does*, as in (i), which was spontaneously uttered by Roberts in response to the question 'Do you know this song?':

i. How's it go?

Here, the form of the main verb *go* clearly indicates that 's corresponds to non-reduced *does*. Likewise, for Roberts examples like (ii) are fully acceptable. In fact, we find examples of this kind in the contemporary British writer Robert Galbraith's (aka J. K. Rowling) (2023) *The Running Grave* (e.g. page 22).

ii. Where's he live?

alternating with HAVE and/or BE, with unaccusatives, in passives and in copular constructions, as shown in (23):¹⁵

- (23) a. *Sevə natə / partutə.* (unaccusative)
 I-SEVA born / left
 ‘I was born / left.’
 b. *Sevə statə vistə.* (passive)
 I-SEVA been seen
 ‘I had been seen.’
 c. *Karlə sevə bbellə.* (copular)
 Carlo SEVA handsome
 ‘Carlo was handsome.’

Furthermore, according to Cennamo (2010:220), in the dialect of San Benedetto del Tronto (spoken in Ascoli Piceno), *seva* can also appear as the auxiliary with unergatives and transitives, as in (24):

- (24) *Səvə dormit / viftə.* (Cennamo 2010:220)
 I-SEVA slept / seen.
 ‘I had slept/seen.’

The *-eva* part of *seva* appears to be derived from the imperfect of HAVE, which is the HAVE auxiliary for the pluperfect in many varieties, including Standard Italian (cf. *Gianni aveva mangiato la mela* ‘Gianni had eaten the apple’). Cennamo (2010:220-3) argues that *seva* is indeed a form of HAVE “with the incorporation of the initial consonant (*s-*) of the present indicative of *be*” (220). We treat *seva* as a neutralised auxiliary resulting from *s*-incorporation into the original HAVE forms, which led to an optional LOC feature being associated with this form. Our proposal for the diachronic development of *seva* is thus as follows:

- (25) [*eva*/əvə_[+LOC] [*s*_[-LOC] [...]] > [*s-eva*/əvə_[±LOC] [[...]]]

¹⁵ These examples are from the Arzano dialect of Campania (Cennamo 2010:213-5).

Thus, Italo-Romance *seva/sɔvə* emerges as a neutralised HAVE/BE auxiliary, like English *ain't*, although by a different diachronic route. Neutralisation is once again captured by the presence of the optional LOC feature.

2.5. Conclusion

We see the importance of the LOC feature both in the analysis of *ain't* and in our discussion of Southern Italo-Romance *seva/sɔvə*. As we have observed, both cases of auxiliary neutralisation involve an optional LOC feature, which arose diachronically in different ways. In the next section, we will broaden the discussion and investigate the relationship between locative and possessive constructions in more general terms, following the influential proposals of Freeze (1992) and Den Dikken (1995).

3. The analysis of locatives and existentials

3.1. Placing the analysis in a wider context

Here we will try to set the analyses of the neutralised auxiliaries just given in a broader theoretical context, which also sheds light on the diachronic processes involved, particularly in regard to labelling.

Adopting and adapting the proposals in Freeze (1992) and Den Dikken (1995), we assume the following structure for locative/possession constructions (see also note 6):

(26) [... [_{VP} BE [_{LocP} LOC [_{XP} Location/Possessor [X Theme]]]]]

Here BE in *v* takes a *LocP* complement, whose head LOC takes an *XP* complement, whose Specifier is a Location/Possessor argument, and whose complement is a Theme. The categorial identity of *XP* is variable.

In fact, *XP* is unstable for labelling, as in Chomsky (2013, 2015) and so, if no other repair strategy is available, the Location/Possessor argument raises out of *XP*. This happens, for example, where LOC incorporates to BE, giving rise to a HAVE auxiliary. This is shown in (27):

(27) [Location/Possessor [_{VP} BE+LOC = HAVE [_{LocP} (LOC) [_{XP} (Location/Possessor) [X Theme]]]]]

This structure thus gives rise to possessive and locative constructions with HAVE, as in basic possessive constructions in English (here we are assuming *have/be* raising, as first proposed by Emonds 1978):

(28) [John [_T has [_{VP} (has) [_{LocP} (LOC) [_{XP} (John) [X a book]]]]]]

Where LOC is realized as a Preposition or an abstract oblique Case-assigner, the Location/Possessor argument is Case-licensed by LOC. Following Saito (2016), we take it that Case-licensed arguments are unable to provide a label. Therefore XP is stably labelled as X. This gives rise to dative (or other oblique) marked possessors. Since LOC does not incorporate here, auxiliary BE results. So, we have (29):

(29) [... [_{VP} BE [_{LocP} LOC-P [_{XP} Location/Possessor_{OBL/DAT} [X Theme]]]]]]

The structure in (29) corresponds to the *mihi est* type of possessive, shown in (30):

(30) [*mihi*_{DAT} [_{VP} *est* [_{LocP} LOC-P [_{XP} (*mihi*) [X *librum*]]]]]]¹⁶

Now let us look again at the paradigm in (4), repeated here, in the light of (26):

- (4)a. There_φ are_φ students in the room.
- b. There_{LOC} 's_{LOC} students in the room.
- c. *There_{LOC} is students in the room.

In all these cases we can treat *there* as the Locative argument first-merged in Spec,XP but not in the complement of *have* (see note 6). Given the instability of this structure, *there* raises in all three examples. In (4a), the LOC head does not raise, and there is simple φ-agreement at the root level as we said in Section 2, giving rise to a φ label for the root. In (4b), LOC raises to BE; in this sense, then, this kind of existential is comparable to the French and Spanish ones in (2) and (12), as we said above. We can assume that *there* has an intrinsic (interpretable) LOC feature; here this feature is crucial for labelling the root, as we saw. In (4a), the feature is present but plays no role in labelling the root. In (4c), we see that LOC doesn't raise; as we said

¹⁶ *Mihi* may be in the left periphery, given the very active left periphery of Latin; see Ledgeway (2012).

in Section 2, in the absence of associate agreement *is* spells out default 3SG features which cannot agree with *there*. Although *there* has its intrinsic LOC feature, there is no shared feature between the Specifier and the head and therefore the root fails to be labelled. In all three cases *students in the room* may correspond to a small clause in the complement of X.

Now consider the following examples with *ain't* (see also (12)):

- (31) a. There ain't no students in the room.
b. John ain't got no book.

In (31a), *there* raises from Spec,XP as just described. As we pointed out in relation to (13) above, *ain't* has a LOC-feature and so there is LOC-agreement with *there* and the root is labelled as LOC. There is no possibility of ϕ -agreement, since *ain't* lacks ϕ -features. In (31b) the situation is very similar: clearly the possessor *John* raises from Spec,XP; again, the root is labelled LOC and there is no ϕ -agreement. Here *got* occupies X, making XP a Participial (Prt) Phrase.

Next, consider how we can extend the analysis of Locative/Possessor constructions to auxiliary selection more generally. In these cases, XP is always PrtP, whose external argument can bear any external-argument role rather than being restricted to Locative/Possessor; this is connected to the grammaticalization of BE as a “perfect auxiliary” as described in Roberts (2013), a matter we will not go into here. So, we have the variant of (26) in (32):

- (32) [... [vP BE [LocP LOC [PrtP EA [Prt (Theme)]]]]]

In the canonical Standard Italian-style auxiliary-selection system, LOC raises to BE, giving HAVE here, and the EA raises to the Specifier of the root (where standard ϕ -agreement takes place, labelling the root in the standard way). In unaccusatives, neither LOC nor the EA is present, so the auxiliary is BE.

In the Italo-Romance varieties where *seva* appears only where there is no EA, such as the Arzano variety seen in (23), the LOC head has disappeared and the auxiliary synchronically optionally has an intrinsic LOC feature. On the other hand, in the varieties where the neutralised auxiliary appears in the context of an EA, as in the dialect of San Benedetto del Tronto in (24), LOC-incorporation to BE as shown in (25) operates synchronically, with the result that *seva* optionally has a +LOC feature, as shown there.

It is tempting to speculate that in passives, LOC could correspond to *by*, or at least function as the head licensing a *by*-phrase in Spec,XP. This would imply that the *by*-phrase, as the external argument (EA), does not raise and that LOC does not raise either, giving rise to the BE auxiliary in passives. Effectively, this analysis would assimilate the active-passive alternation to the alternation between HAVE-possessives and *mihi est* possessives. However, the idea that the *by*-phrase corresponds to the active external argument is controversial; see for example Roberts (1987) and Kallulli (2007) for different views on the status of the external argument in short verbal passives. For this reason, we leave this question open here.

3.2. A diachronic generalisation

In his classic discussion of the different kinds of possessor constructions, Benveniste (1971: 170) points out that “the [diachronic DK, IR] development is from ‘*mihi est*’ to ‘*habeo*’ and not the reverse”. As far as we are aware, this is a correct observation. Assuming it is accurate, then, we should try to find an account for it.

In this context, consider again the *habeo* configuration (after (27)), and the *mihi est* configuration (after (29)). Both (27) and (29) are repeated here:

(27) [Location/Possessor [_{VP} BE+LOC = *habeo* [_{LocP} (LOC) [_{XP} (Location/Possessor) [X Theme]]]]]

(29) [TP ... [_{VP} BE [_{LocP} LOC-P [_{XP} *mihi*_{OBL} [X Theme]]]]]

We see that in (29) the LOC feature is on the Locative/Possessor *mihi*. As an oblique, it therefore does not contribute to labelling, following Saito (2016). In (27), on the other hand, LOC is associated with T after raising of *habeo*. In this position, in conjunction with the right kind of element merged to it, LOC can contribute to a label, as we have seen. So, we could construe the propensity for change just in the direction described by Benveniste as favouring the use of the formal feature LOC as a label.

There is a further issue here: the diachronic change from *mihi est* to *habeo* is also an instance of a shift from dependent- to head-marking in the sense of Nichols (1986); see also Hallman (2022). Still comparing (27) and (29), we see that the head-marking configuration is the one where the head directly contributes to labelling while the dependent marking configuration appears to prevent this, requiring some independent operation for labelling (in fact, it is unclear

what labels the root in the *mihi est* construction; in a language like Latin, this is presumably the “strong” ϕ -features of T since Latin is a null-subject language, see again Chomsky 2015). Nichols observed a correlation between head-marking and head-initial order and dependent-marking and head-final order (Nichols 1986). So, we observe the following correlations:

- (33) a. Head-marking, head-initial orders, head-labelling.
 b. Dependent-marking, head-final orders, “other” labelling.

It is very tempting to connect “other” labelling to the triggering of XP-movement operations deriving head-final orders in an asymmetric linearisation system of the kind proposed in Kayne (1994); this idea is developed in Roberts (2019: 167f.), although the assumptions made there are somewhat idiosyncratic. Across a range of older Indo-European languages, including of course the transition from Latin to Romance, we observe change from basic head-final typology to head-initial typology. It may be that Benveniste’s generalisation is part of that wider set of changes, suggesting a very deep labelling-based parameter governing the properties in (33). However, we will leave these more wide-ranging speculative matters for future research.

4. Conclusions

Here we have made several proposals. First, we have proposed that auxiliaries which are underspecified for being HAVE or BE, such as English *ain’t* and Southern Italo-Romance *seva/sɔva* have an optional LOC feature. By comparing the development of the two different cases of neutralised auxiliaries, we saw that the convergence of HAVE and BE can arise in different ways (phonologically or morphologically). This is consistent with Postma’s (1993:31) observation that “the defective nature of BE does not stem from the lexicon, but must have an inherent morpho-syntactic origin”. As we saw in Section 3, our account of the neutralised auxiliaries can readily be integrated into a version of the Freeze approach to locatives and existentials. In fact, we have seen support for Freeze’s (1992:576) claim that “the ‘have’ predication *is* the existential”.

Building on this, we can account for the following statement in Freeze (1992:580): “[g]iven the syntactic similarity of the existential and the ‘have’ predication, we would predict morphological similarities as well. In fact, [...] in many languages the existential and the ‘have’ predication share the same copula form, one that is distinct from that of the predicative locative and of other copular constructions. Other examples of such a copula are Mandarin Chinese

you, Shanghainese *yu*, Hebrew *yes*, Navajo *holg*, French *avoir*, Portuguese *ter*, [...] Modern Greek *echei*, Quechua *tiya*, Tagalog *may*, Trukese *mei*, Yosondua Mixtec (Oto-Mangue) *yo*, and Turkish *var*.”

To this list we can add SE 's as in (1c) and non-standard English *ain't*.

On a more theoretical level, we have seen that labelling preferences, more specifically an apparent preference for head-labelling, can drive syntactic change. This conclusion converges with those of Dadan (2019) and van Gelderen (2022), who argue for exactly this kind of syntactic change in the case of wh-phrases being reanalysed as C heads (see in particular Dadan on this point). In fact, van Gelderen derives her earlier “Head Preference Principle” (van Gelderen 2004), which accounts for many cases of grammaticalisation, from the labelling algorithm.

References

- Benveniste, Émile. 1960. “Être” et “Avoir” dans leurs fonctions linguistiques. *Bulletin de la Société de Linguistique* LV. (Reprinted in *Problèmes de Linguistique Général*, Gallimard, 1966, 187-207. Paris.)
- Benveniste, Émile. 1971. *Problems in General Linguistics*. University of Miami Press.
- Boneh, Nora and Ivy Sichel. 2010. Deconstructing possession. *Natural Language and Linguistic Theory* 28:1-40.
- Broadbent, Judith. 2009. The **amn't* gap: The view from West Yorkshire. *Journal of Linguistics* 45: 251–284
- Cennamo, Michela. 2010. Perfective auxiliaries in the pluperfect in some southern Italian dialects. In D'Alessandro, R., A. Ledgeway and I. Roberts (eds.) *Syntactic Variation: The Dialects of Italy*. CUP: 210-224.
- Cheshire, Jenny. 1982. *Variation in an English Dialect: A Case Study*. Cambridge University Press.
- Chomsky, Noam. 2013. Problems of Projection. *Lingua* 130:33-49.
- Chomsky, Noam. 2015. Problems of Projection: Extensions. In Di Domenico, E., C. Hamann and S. Matteini (eds.) *Structures, Strategies and Beyond: Studies in Honour of Adriana Belletti* 1-16. Benjamins.
- Close, Joanne. 2004. *English Auxiliaries: A Syntactic Study of Contraction and Variation*. PhD dissertation, University of York.
- Dadan, Marcin. 2019. *Head Labeling Preference and Language Change*. PhD dissertation, University of Connecticut.

- Den Dikken, M. 1995. *Particles*. Oxford: Oxford University Press.
- Emonds, Joseph. 1978. The complex V-V' in French. *Linguistic Inquiry* 9(2):151-175.
- Fisher, Sabriyah. 1998. *Variation and Change in Past Tense Negation in African American English*. PhD dissertation, University of Pennsylvania.
- Freeze, Ray. 1992. Existentials and other locatives. *Language* 68: 553-595.
- Gelderen, Elly van. 2022. *Third Factors in Language Variation and Change*. Cambridge University Press.
- Gelderen, Elly van. 2004. *Grammaticalization as Economy*. Amsterdam: John Benjamins.
- Hallman, Peter. 2022. Head and dependent marking in clausal possession. *Linguistic Inquiry* 53 (3): 551–570.
- Hazout, Ilan. 2004. The syntax of existential constructions. *Linguistic Inquiry* 35: 393-430.
- Henry, Allison. 1995. *Belfast English and Standard English*. Oxford University Press.
- Jenkins, Lyle. 1975. *The English Existential*. Tübingen: Niemeyer.
- Kallulli, Dalina. 2007. Rethinking the passive/anticausative distinction. *Linguistic Inquiry* 38(4):770-780.
- Kallulli, Dalina. 2008. There is Secondary Predication in *There*-Existentials. In Charles B. Chang and Hannah J. Haynie (eds) *Proceedings of the 26th West Coast Conference on Formal Linguistics* 279-287. Somerville, MA: Cascadilla Proceedings Project.
- Kayne, Richard. 1993. Toward a modular theory of auxiliary selection. *Studia Linguistica* 47: 3-31. [Reprinted in Kayne 2000 *Parameters and Universals*, Oxford University Press].
- Landau, Idan. 2010. *The Locative Nature of Experiencers*. MIT Press.
- Lass, Roger. 1999. Phonology and Morphology. In R. Lass (ed) *The Cambridge History of the English Language, Volume III 1476-1776*. Cambridge University Press.
- Ledgeway, Adam. 2012. *From Latin to Romance: Morphosyntactic Typology and Change*. Oxford University Press.
- Levinson, Lisa. 2011. Possessive WITH in Germanic: HAVE and the Role of P. *Syntax* 14(4):355-393.
- Loporcaro, Michele. 2016. Auxiliary selection and participial agreement. In A. Ledgeway and M. Maiden (eds) *Oxford Guide to the Romance Languages*. Oxford University Press: 802-20.
- Manzini, Rita and Leonardo Savoia. 2005. *I dialetti italiani e romance: morfosintassi generativa*, vols 1-3. Alessandria: Edizioni dell'Orso.

- Marantz, Alec. 1991. Case and licensing. In German Westphal, Benjamin Ao and Hee-Rahk Chae (eds.) *Proceedings of the 8th Eastern States Conference on Linguistics* (ESCOL 8) 234–253. Ithaca, NY: CLC Publications.
- Nichols, Johanna. 1986. Head-marking and dependent-marking grammar. *Language* 62:56-119.
- Postma, Gertjan. 1993. The Syntax of the morphological defectivity of the BE. *HIL manuscripts* 3: 31-67.
- Roberts, Ian. 1987. *The Representation of Implicit and Dethematized Subjects*. Dordrecht: Foris.
- Roberts, Ian. 1993. *Verbs and Diachronic Syntax*. Springer.
- Roberts, Ian. 2010. *Agreement and Head Movement. Clitics, Incorporation, and Defective Goals*. MIT Press.
- Roberts, Ian. 2013. Some speculations on the development of the Romance periphrastic perfect. *Revue Roumaine de Linguistique* 58: 1-30.
- Roberts, Ian. 2019. *Parameter Hierarchies and Universal Grammar*. Oxford: Oxford University Press.
- Saito, Mamoru. 2016. (A) case for labeling: labeling in languages without phi-feature agreement. *The Linguistic Review* 33: 129-75.
- Thoms, Gary, David Adger, Caroline Heycock, and Jennifer Smith. 2019. Syntactic variation and Auxiliary contraction: the surprising case of Scots. *Language* 95(3): 421-455.
- Thoms, Gary, David Adger, Caroline Heycock, E Jamieson, and Jennifer Smith. 2023a. Explaining microvariation using the Tolerance Principle: plugging the *amn't* gap. Ms., NYU/QMUL/Edinburgh/York/Glasgow.
- Thoms, Gary, David Adger, Caroline Heycock, E Jamieson, and Jennifer Smith. 2023b. *Have-raising and the structure of predicative possession*. Ms., NYU/QMUL/Edinburgh/York/Glasgow.
- Williams, Edwin. 1994. *Thematic Structure in Syntax*. Cambridge MA, MIT Press.